

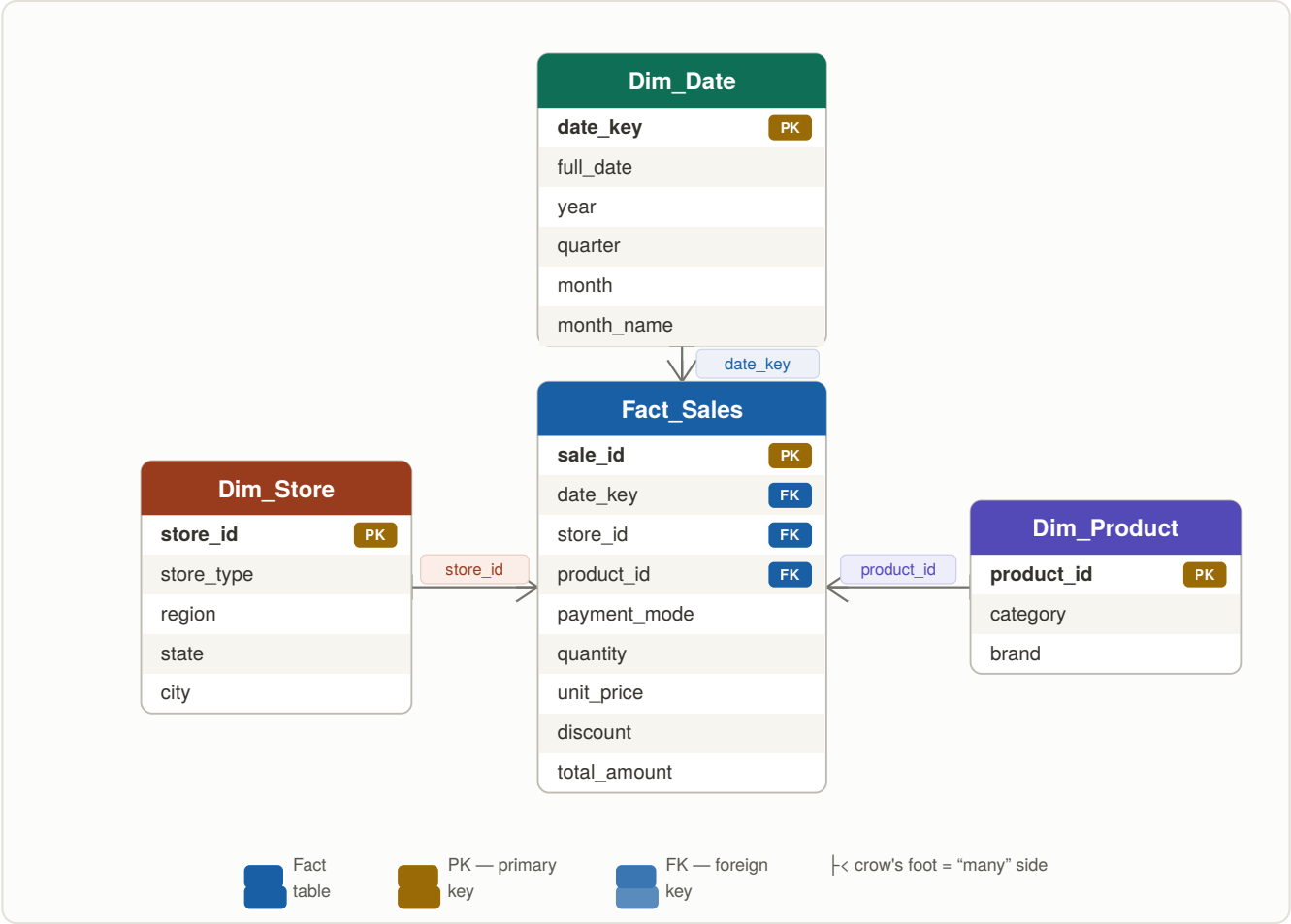
Sales Data Model — Star Schema

Workbook: GenFlow_Validation • Dimensional model for review

Status: Draft for review
Grain: One row per sale
Tables: 1 fact • 3 dimensions

This model organises the sales data as a **star schema**: a central fact table holding the measurable events, surrounded by dimension tables holding descriptive context. Each foreign key in Fact_Sales joins to the primary key of its dimension. All relationships are **one-to-many** (one dimension row relates to many fact rows).

Data model diagram



Relationships (modelling logic)

Fact column (FK)	Joins to	Dimension key (PK)	Cardinality
Fact_Sales.date_key	→	Dim_Date.date_key	many → one
Fact_Sales.store_id	→	Dim_Store.store_id	many → one
Fact_Sales.product_id	→	Dim_Product.product_id	many → one

How the model is built

- Load all four sheets as separate tables — keep Fact_Sales as the fact; do not flatten the dimensions into it.
- Mark each dimension id (date_key, store_id, product_id) as its primary key.
- Create the three relationships above as single-direction, one-to-many joins.
- Build measures on the fact table (sum of total_amount, quantity) and slice by dimension attributes (region, brand, category, year, quarter).

Validation checklist

- ☐ sale_id is unique — confirms the one-row-per-sale grain.

- ☐ Dimension keys are unique in each dimension (no duplicate `store_id` / `product_id` / `date_key`).
- ☐ Referential integrity: every FK value in `Fact_Sales` exists in its dimension (no orphan keys).
- ☐ Confirm derivation of `total_amount` — in the sample it does not equal `quantity × unit_price - discount`, so the measure logic needs sign-off.
- ☐ Decide on surrogate vs natural keys — ids are currently reused business keys.

Reviewer sign-off

Reviewer name

Date

Decision: ☐ Approved ☐ Approved with changes ☐ Changes required

Comments